

Exponent Rules with Reasons



Grade 8 • Mathematics
NC.8.EE.1

Generate equivalent exponential expressions.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

Simplify $2^7 \div 2^5$ as one power and evaluate it. Explain by canceling equal factors.

Cancel 5 nonzero factors from numerator and denominator. $7 - 5 = 2$ factors remain.



Build a second model together

Simplify $(4^5)(4^4)$ as one power and evaluate it. Explain using repeated factors.

Discuss: What should the first step find?



Your turn • 1

Simplify $2^5 \div 2^3$ as one power and evaluate it. Explain by canceling equal factors.

Represent → solve → explain → check



Your turn • 2

Simplify $(2^5)(2^2)$ as one power and evaluate it. Explain using repeated factors.

Represent → solve → explain → check



Your turn • 3

Simplify $5^4 \div 5^2$ as one power and evaluate it. Explain by canceling equal factors.

Represent → solve → explain → check



Your turn • 4

Simplify $(5^4)(5^1)$ as one power and evaluate it. Explain using repeated factors.

Represent → solve → explain → check



Exit question • Explain independently

Simplify $2^6 \div 2^3$ as one power and evaluate it. Explain by canceling equal factors.

Include a check. What would you teach someone else?



Teacher review and next steps

1. $2^2 = 4$

2. $2^7 = 128$

3. $5^2 = 25$

4. $5^5 = 3125$

Exit: $2^3 = 8$

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

